

HW 4

1a. Python

b. Python

c. Python

d. As shown by e, the proportion of variance explained by components, also known as the PCA is calculated through the SVD of the centered data matrix

e. 2: 28.51%

5: 54.50%

10: 73.82%

20: 89.43%

f. Python

g. Python

h. The original image looks the most like a "5", the noisy image does not really show any signs of being any specific image, $k=5$ is the noisiest with a slight pattern of "5", $k=10$ is in the middle with some noise and as sign of the original image, $k=20$ has the ~~best~~ noise and is very close to the original image. The choice of k affects the reconstruction quality because a lower k will have more noise and broad features and the noise will decrease as k increases, but eventually a too high k will reintroduce noise.

HW 4

4a. Python

b. pdf $f(x, \lambda) = \lambda e^{-\lambda x}$ $x > 0, \lambda > 0$

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

c. $l(\lambda) = n \ln(\lambda) - \lambda \sum_{i=1}^n x_i \ln(e)$
 $= n \ln(\lambda) - \lambda \sum_{i=1}^n x_i$

d. $\frac{d}{d\lambda} l(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0$ $\lambda \left(\frac{n}{\lambda} = \sum_{i=1}^n x_i \right)$
 $n = \lambda \cdot \sum_{i=1}^n x_i$ $\lambda = \frac{n}{\sum_{i=1}^n x_i}$ $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

$$\hat{\lambda}_{MLE} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}$$

e. $\hat{\lambda}_{MLE} = 0.3335$

f. $E(x) = 1/\lambda$ $E(x) = 1/\hat{\lambda}$ $E(x) = 2.9983$

g. Python

h. sample mean = 2.9983

model mean = $E(x) = 2.9983$

These two means are exactly the same.

i. The exponential model appears to be an okay choice for this dataset. The data is skewed right like the exponential distribution, but the data is not consistently decreasing from 0 like the exponential. Thus, the exponential meets the skewed right criteria, but does not fit the model perfectly.

HW 4

5a. Python

b. 344 rows, 8 columns (344, 8)

Missing Values:

Species: 0

island: 0

bill-length-mm: 2

bill-depth-mm: 2

flipper-length-mm: 2

body-mass-g: 2

sex: 11

year: 0

c. Python

d. Python Now (333, 7)

e. Python Now (333, 9)

f. Python Now (286, 9)

g. Python

h. Python

i. Based on the boxplots, the Adèle distribution is roughly symmetric with a median at 3.75, an IQR of about 0.5, and a range of 2. The Chinstrap distribution is roughly symmetric with a median at 3.75, an IQR of about 0.5, and a range of about 1.5. It also contains 2 outliers. The Geico distribution is roughly symmetric with a median at 5, an IQR of about 0.75, and a range of 2.25. The main differences are that the Geico species has the largest range and IQR of body mass, followed by the Adèle, then the Chinstrap. The Geico also has the largest median body mass. Lastly, the Chinstrap species contains two outlier body masses, one above and one below.